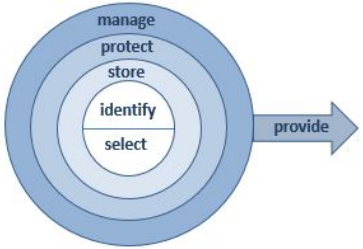
Best Practices for Biomedical Research Data Management

Module 8: Curation & Preservation for Data Digital Preservation

Nancy McGovern, PhD, MIT



This video will expand on the topics defined earlier, including: digital curation, digital preservation, data curation, and long-term access. These terms are vital to understanding how digital content is preserved, curated and made accessible for the the long-term.

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Digital Preservation

\+ Data **Curation**

Digital Curation

Digital Curation Centre, 2004

Digital Preservation

"the active management of digital content over time to ensure ongoing access" (NDIIPP*)

- Encourage quality creation by producers
- Document actions taken over the life of digital objects
- Ensure access over time
  - Handshakes across generations of technology
  - Proven technologies for preservation to contemporary for access

* National Digital Information Infrastructure and Preservation Program Library of Congress

Long-term Access

Preservation makes long-term access possible…

Preservation	**vs.**	**Access**
proven	<- technologies ->	cutting edge
accumulate	<- metadata ->	relevant now
access over time	<- purpose ->	access now
future users	<- focus ->	current users

Data

Can be defined narrowly –

data is a result of scientific research

And broadly –

data refers to digital information of any kind

Or, data = digital content

Research data

"The data, records, files or other evidence, irrespective of their content or form (e.g. in print, digital, physical or other forms), that comprise research observations, findings or outcomes, including primary materials and analysed data."

Australian National Data Service

Curation

Curation may refer to:

- preservation and maintenance of digital assets

Curation may also refer to:

- Work performed by a curator (initially in a museum context)
- Archiving, historical record keeping
- Evidence Management, indexing and cataloguing evidence
- Cultural heritage management, protection of cultural sites

Other uses: healing, religion …

Source: wikipedia

Data Curation

"active and on-going management of data through its life cycle of interest and usefulness to scholarship, science, and education"

enables discovery, ensures quality, adds value, and provide for re-use over time [UIUC]

- Predates the digital community
- Value-added steps by curators to enhance utility
- Intersection of research and curation

Digital Curation

"maintaining and adding value to a trusted body of digital information for future and current use"

- active management and appraisal over entire life cycle
- builds upon underlying concepts of digital preservation
- emphasizes opportunities for adding value through annotation and continuing resource management
- Preservation is a curation activity - both are concerned with managing digital resources with no significant (or only controlled) changes over time

Source: JISC

Life Cycle Management

- Connect the dots now and into the future
  - Link actions across units, **workflows**, content
  - Document ‘as is’ and define ‘to be’ workflows
  - Handshakes between stages, over time
- Document conformance with good practice (standards and community norms)
- Align practice for all content types
- Develop and maintain skills